

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – IV
PAPER –2
TEST DATE: 31-07-2022

ANSWERS, HINTS & SOLUTIONS

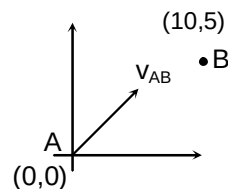
Physics

PART – I

Section – A

1. AC

Sol. $\vec{v}_{AB} = (3-a)\hat{i} + (3-b)\hat{j}$
 $\vec{a}_{AB} = \vec{0}$
 $(3-a) \times 2 = 10$ and $(3-b) \times 2 = 5$
 $a = -2$ and $b = \frac{1}{2}$



2. ACD

Sol. $\vec{F}_m = q(\vec{v} \times \vec{B})$ and apply Newton's second law of motion.

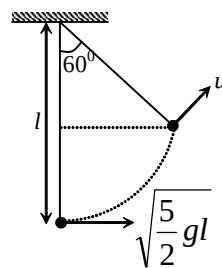
3. ABCD

Sol. $T - mg \cos 60^\circ = \frac{mv^2}{l}$... (i)

$\frac{1}{2} m \left(\sqrt{\frac{5}{2} gl} \right)^2 - mg \frac{l}{2} = \frac{1}{2} mv^2$... (ii)

$v = \sqrt{\frac{3}{2} gl}$ and $T = 2mg$

Net force is in horizontal direction when $\theta = 60^\circ$



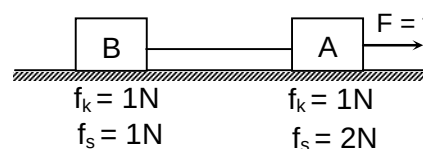
4. ABC

Sol. When $t = 3\text{s}$ block just about to move and acceleration of block given by $a = \frac{t-2}{1} \quad t > 3$

$$\int_0^v dv = \int_3^{10} (t-2) dt$$

$$v = \frac{t^2}{2} - 2t \Big|_3^{10} = (50 - 20) - \left(\frac{9}{2} - 6 \right)$$

$$= 30 + 1.5 = 31.5 \text{ m/s}$$



5. BCD

Sol. Sign between x and t is $\oplus$ hence wave is propagating in negative x direction.

$$\text{wave velocity} = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{3}{1}$$

Max value of y is amplitude, y has maximum value when denominator is least i.e.

when $(x + 3t)^2$ is zero

$$y = 3\text{m}$$

$$\text{shape of pulse is given by } y = \frac{6}{2 + x^2} \quad (t = 0)$$

shape of pulse is symmetrical out y -axis

6. BC

$$\text{Sol. } \tan \alpha = - [\text{slope at } (P_1, V_1)] = \frac{P_1}{V_1}$$

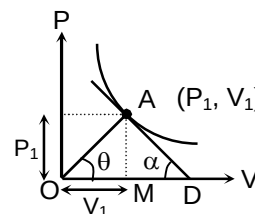
$$\tan \theta = \frac{P_1}{V_1}$$

$$\therefore \alpha = \theta$$

$$OM = MD = V_1$$

$$\therefore OD = 2V_1$$

$$\text{Area of triangle } AOD = \frac{1}{2} (AM \times OD) = \frac{1}{2} (P_1 \times 2V_1) = P_1 V_1 = nRT$$



7. D

Sol. The net force acting on the system is zero. So, linear momentum of the system remains constant.

8. B

Sol. Linear momentum

$$mv_0 = 2mv,$$

$$v_1 = \frac{v_0}{2}.$$

9. C

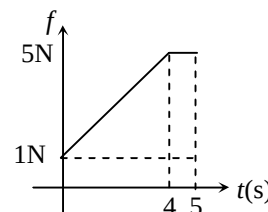
Sol. $T_1 = \frac{dm}{dt}v + \frac{dm}{dt} \times tg = 1 + 5 = 6 \text{ N}$

10. A

Sol. $f_{\max} = 5 \text{ N}$

At $t = 0$ $f_r = T_1 = \frac{dm}{dt}v = 1 \text{ N}$

Impulse of friction force = area under $f-t$ curve = 17 N-s



Section – B

11. 6

Sol. Energy in $2n^{\text{th}}$ state will be $= -\frac{13.6Z^2}{4n^2} \text{ eV}$

Max energy of photon $= 13.6Z^2 \left(1 - \frac{1}{4n^2}\right) = 204 \dots(1)$

Energy of photon when it makes transition to x in state will be

$13.6Z^2 \left(\frac{1}{n^2} - \frac{1}{4n^2}\right) = 40.8$

.....(2)

Dividing equation (1) and (2)

$$\frac{4n^2 - 1}{3} = 5$$

$$\Rightarrow n^2 = 4 \quad n = 2$$

Putting of $n = 2$ in equation (2)

$$13.6Z^2 \left(\frac{3}{4 \times 4}\right) = 40.8$$

$$Z^2 = \frac{40.8 \times 16}{3 \times 13.6}$$

$$Z = 4$$

12. 1

Sol. $v_0 \leq \frac{gR}{3} (7 \cos \alpha - 4)$

13. 1

Sol. $a = \omega^2 A$

$f_1 = (1)\omega^2 A \leq 6 \dots(1)$

$f_2 = 3\omega^2 A \leq 12$

$\omega^2 A \leq 4 \dots(2)$

$\Rightarrow \omega^2 A = 4$

$$A = \frac{4}{\left(\frac{k}{m}\right)} = \frac{4 \times 6}{24} = 1$$

Section – C

14. 0.61

15. 33.77

Sol. (Combined solution 14-15)

Mass per unit length of the wire = 0.1 kg/m.

At any instant, tension in the string is $(2 + 2.25 t)$ g N

$$v = \sqrt{\frac{(2 + 2.25t)g}{0.1}} ; \int_0^{10} dx = 10 \int_0^t \sqrt{2 + 2.25t} dt$$

$$t = 0.612 \text{ s}$$

$$T = (2 + 2.25 \times 0.612) \times 10 = 33.77 \text{ N}$$

16. 17.50

Sol. From symmetry $V_1 = V_4$ and $V_2 = V_3$

$$\therefore V_1 + V_2 = 30$$

$$V_1 - V_2 = 5 \text{ (given)}$$

$$\therefore V_1 = 17.5 \text{ V}$$

$$V_2 = 12.5 \text{ V}$$

$$\therefore V_4 = V_1 = 17.5 \text{ V}$$

17. 2.34

Sol. $C = \frac{C_1 V_1}{V_2} = \frac{35}{12.5} \mu\text{F}$

$$\therefore \text{Required capacitance} = 2 \left(\frac{C_1 C_2}{C_1 + C_2} \right)$$

$$= \frac{2 \times 2 \times \frac{35}{12.5}}{2 + \frac{35}{12.5}} = \frac{140}{25 + 35} = \frac{7}{3} = 2.34 \mu\text{F}$$

18. 4.00

Sol. For 30 kg block

$$210 - kx = 30a$$

$$a = \frac{210 - 90}{30} = 4 \text{ m/s}^2$$

19. 0.90

Sol. For 15 kg block

$$Kx = ma = 15 \times 6$$

$$x = \frac{90}{100} = 0.9 \text{ m}$$

Chemistry**PART – II****Section – A**

20. ABD

Sol. O_2^+ contains unpaired electron in π -antibonding M.O and N_2^+ in σ_{2p} orbital N_2^{2-} contains unpaired electron in π -antibonding M.OS.

21. ABC

Sol. $\text{Rate} = k[X][Y]^0 \Rightarrow 2 \times 10^{-8} = k[0.1][0.1]^0 \Rightarrow k = 2 \times 10^{-7} \text{s}^{-1}$.

22. ABD

Sol. When Cl is directly attached to benzene, +R effect of chlorine is observed. Which increases the bond order of C – Cl bond.

23. BCD

Sol. Elimination takes place in option(A).

24. AC

Sol. Azeotropic mixture has fixed composition.

25. ACD

Sol. $a = 2(r_{Na^+} + r_{Cl^-}) = 2(95 + 181) = 552 \text{ pm}$

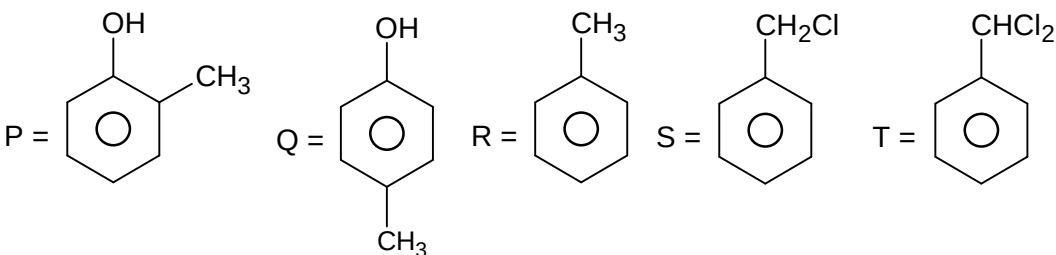
26. C

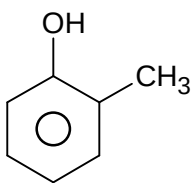
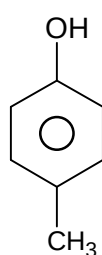
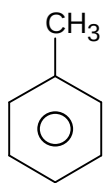
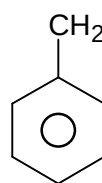
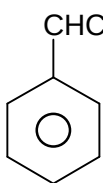
Sol. $D = \frac{\text{mol.mass of } SO_3}{2} = \frac{80}{2} = 40$

27. C

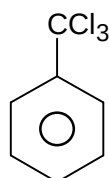
Sol. $\alpha = \frac{40 - 30}{(1.5 - 1)30} = \frac{10}{15} = 0.66$

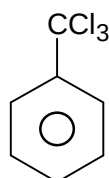
28. D

Sol.


 P = 
 Q = 
 R = 
 S = 
 T = 

29. B

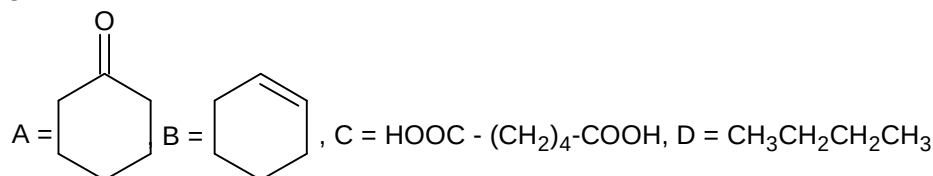
Sol.


 U = 

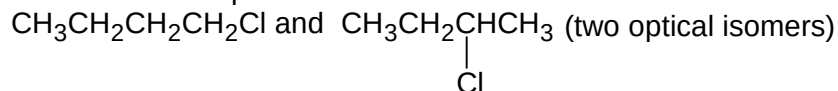
Section – B

30. 3

Sol.



The monochloro product are



31. 4

 Sol. $A = \text{CaO}, B = \text{Ca}(\text{OH})_2, C = \text{CaOCl}_2$

32. 8

 Sol. $\Delta T_f = K_f m$

$$\text{or } 0 - (-5) = 1.86 \times 1 \times \frac{1000}{W}$$

$$\therefore W = 372 \text{ g}$$

$$\therefore \text{mass of ice} = 380 - 372 = 8 \text{ g}$$

Section – C

33. 0.50

34. 1.00

Sol. (Q.33 & 34):

$$\text{Moles of } \text{Na}_2\text{CO}_3 = \frac{10.6}{106} = 0.1$$

$$\text{Moles of } \text{NaOH} = \frac{4}{40} = 0.1$$

Two moles of HCl is needed for complete reaction of Na_2CO_3 . But in presence of phenolphthalein indicator, one mole of HCl is required. So total moles of HCl required = $0.05(\text{Na}_2\text{CO}_3) + 0.1(\text{NaOH})$
 $= 0.15 \text{ mole}$, Volume of HCl = 300 mL

$$\therefore M = 0.15 \times \frac{1000}{300} = 0.5$$

35. 10.00

36. 9.83

 Sol. (Q.35 & 36): $\text{NaCN} + \text{HNO}_3 \rightleftharpoons \text{NaNO}_3 + \text{HCN}$

Before mixing	500×0.2	500×0.1	0	0
	= 100	= 50	50	50

After mixing	50	0	50	50
--------------	----	---	----	----

$$\text{pH} = \text{p}^{K_a} + \log \frac{[\text{NaCN}]}{[\text{HCN}]} = 10 + \log \frac{50}{50} = 10$$

$$\therefore x = 10$$

After adding of HCl (0.01 mole)

$$\text{pH} = \text{p}^{K_a} + \log \left[\frac{\frac{50}{1000} - 0.01}{\left[\frac{50}{1000} + 0.01 \right]} \right]$$

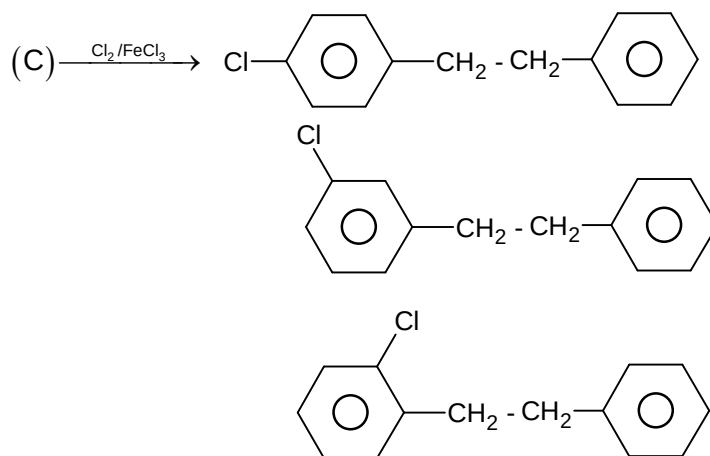
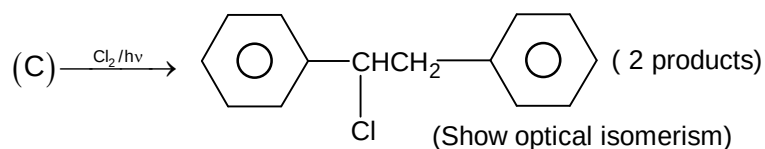
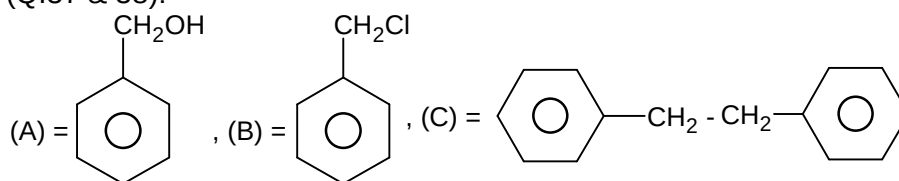
$$= 10 + \log \frac{2}{3} = 10 - 0.17 = 9.83$$

$$\therefore y = 9.83$$

37. 2.00

38. 3.00

Sol. (Q.37 & 38):



Mathematics

PART – III

Section – A

39. AC

Sol. $\therefore f(x) = \begin{vmatrix} n & n+1 & n+2 \\ {}^nP_n & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^nC_n & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$
 $= \begin{vmatrix} n & n+1 & n+2 \\ n! & (n+1)! & (n+2)! \\ 1 & 1 & 1 \end{vmatrix} \quad (\because {}^nP_n = n!, {}^nC_n = 1)$
 Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$
 Then $f(x) = \begin{vmatrix} n! & 1 & 2 \\ n! & n.n! & (n^2 + 3n + 1)n! \\ 1 & 0 & 0 \end{vmatrix}$
 $= \begin{vmatrix} 1 & 2 \\ n.n! & (n^2 + 3n + 1)n! \end{vmatrix}$
 $= n!(n^2 + n + 1)$

40. ABC

Sol. $\therefore |A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix} = 3(-3 + 4) - 2(-3 + 4) + 0 = 1$
 $\therefore \text{adj}(\text{adj } A) = |A|^{3-2} A = A$ and $|\text{adj}(\text{adj } A)| = |A| = 1$
 Also, $|\text{adj } A| = |A|^{3-1} = |A|^2 = 1^2 = 1$

41. BD

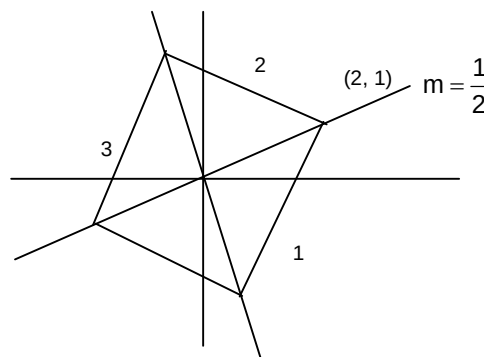
Sol. Since, $\sin \theta + \sqrt{2} \cos \theta = 6x - x^2 - 11$
 $\Rightarrow \sin\left(\theta + \frac{\pi}{3}\right) = \frac{6x - x^2 - 11}{2} \dots\dots\dots(i)$
 $\therefore -1 \leq \sin\left(\theta + \frac{\pi}{3}\right) \leq 1$
 $\Rightarrow -1 \leq \frac{6x - x^2 - 11}{2} \leq 1$
 $\Rightarrow -2 \leq 6x - x^2 - 11 \leq 2$
 $\therefore x = 3$
 From Equation (i), $\sin\left(\theta + \frac{\pi}{3}\right) = -1$

$$\begin{aligned}\theta + \frac{\pi}{3} &= 2n\pi - \frac{\pi}{2} \\ \Rightarrow \theta &= 2n\pi - \frac{5\pi}{6} \\ \text{For } \theta &= \frac{7\pi}{6}, \frac{19\pi}{6} \quad (\because 0 \leq \theta \leq 4\pi) \\ \text{or } (x, \theta) &= \left(3, \frac{7\pi}{6}\right), \left(3, \frac{19\pi}{6}\right)\end{aligned}$$

42. AB

Sol. The point of intersection of diagonal is (0, 0)

and one vertex A (2, 1)

 $\Rightarrow$ Vertex is (-2, -1) vertices B and D are $\pm i(2 + i)$ $= -1 + 2i, 1 - 2i$ $\Rightarrow B(-1, 2)$ and $D(1, -2)$ $\Rightarrow$ Equation of the sides are $x + 3y = \pm 5, 3x - y = \pm 5.$ 

43. ABCD

Sol. As a_{912}, a_{951} and a_{480} are divisible by 3, none of them is prime.For a_{91} , we have

$$a_{91} = \frac{1}{9} \underbrace{(99 \dots 9)}_{91 \text{ times}}$$

$$= \frac{(10^{91} - 1)}{9}$$

$$= \frac{1}{9} \left[(10^7)^{13} - 1 \right]$$

$$= \left[\frac{(10^7)^{13} - 1}{10^7 - 1} \right] \left[\frac{10^7 - 1}{10 - 1} \right]$$

$$= \left[(10^7)^{12} + (10^7)^{11} + \dots + 10^7 + 1 \right] \times [10^6 + 10^5 + \dots + 10 + 1]$$

 $= a_{91}$ is not prime.

44. ABD

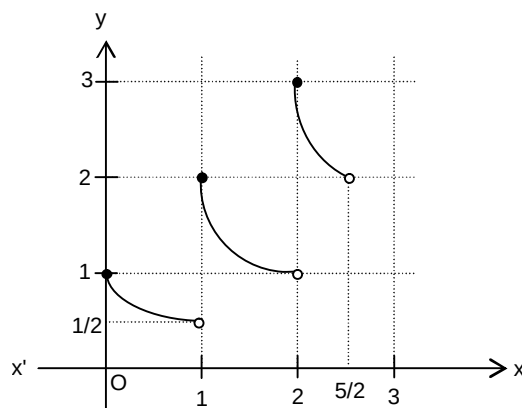
Sol.

$$f(x) = \begin{cases} \frac{1}{1+x}, & 0 \leq x < 1 \\ \frac{2}{x}, & 1 \leq x < 2 \\ \frac{3}{x-1}, & 2 \leq x < \frac{5}{2} \end{cases}$$

Clearly, $f(x)$ is discontinuous and bijective function

$$\lim_{x \rightarrow 1^-} f(x) = \frac{1}{2}, \quad \lim_{x \rightarrow 1^+} f(x) = 2$$

$$\min\left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x)\right) = \frac{1}{2} \neq f(1)$$



45. B

Sol. If $E_n : \frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1$ and eccentricity of E_n is e_n

$$\text{If } a_n > b_n$$

$$\text{Then } b_n^2 = a_n^2(1 - e_n^2) \quad \dots\dots\dots(i)$$

According to question

$$b_n = b_{n-1} \quad \dots\dots\dots(ii)$$

$$\text{and } a_{n-1} = a_n e_n \quad \dots\dots\dots(iii)$$

For ellipse E_{n-1} ,

$$a_{n-1}^2 = b_{n-1}^2(1 - e_{n-1}^2) \quad \dots\dots\dots(iv) \quad (\text{let } b_{n-1} \text{ and } a_{n-1} \text{ is semi major and semi minor axis of } E_{n-1})$$

From equation (i) and (ii),

$$b_{n-1}^2 = a_n^2(1 - e_n^2) \quad \dots\dots\dots(v)$$

Substituting the values of a_{n-1} and b_{n-1}^2 from equation (iii) and (v) in (iv), then

$$\begin{aligned} a_n^2 e_n^2 &= a_n^2(1 - e_n^2)(1 - e_{n-1}^2) \\ \Rightarrow e_n^2 &= (1 - e_n^2)(1 - e_{n-1}^2) \quad \dots\dots\dots(vi) \end{aligned}$$

$\therefore E_n$ is independent of n ,

$$\therefore e_n = e_{n-1} = e \quad (\text{say})$$

$$\Rightarrow e^2 = (1 - e^2)^2$$

$$\Rightarrow e^4 - 3e^2 + 1 = 0$$

$$\begin{aligned} \therefore e^2 &= \frac{3 \pm \sqrt{5}}{2} \\ &= \frac{6 \pm 2\sqrt{5}}{4} \end{aligned}$$

$$= \left(\frac{\sqrt{5} \pm 1}{2} \right)^2$$

$$\therefore e = \frac{\sqrt{5}-1}{2} \quad (\because 0 < e < 1)$$

46. C

Sol. $E_1 = \frac{x^2}{9} + \frac{y^2}{16} = 1$

$$\therefore a_1 = 3, b_1 = 4$$

$$\text{and } 9 = 16(1 - e_1^2)$$

$$\therefore e_1 = \frac{\sqrt{7}}{4}$$

$$\text{From equation (vi), } e_2 = \sqrt{\frac{1 - \frac{7}{16}}{2 - \frac{7}{16}}} = \frac{3}{5}$$

$$\text{and then } e_3 = \sqrt{\frac{1 - \frac{9}{25}}{2 - \frac{9}{25}}} = \frac{4}{\sqrt{41}}$$

$$\text{Also } a_1 = a_2 e_2 \text{ and } b_1 = b_2$$

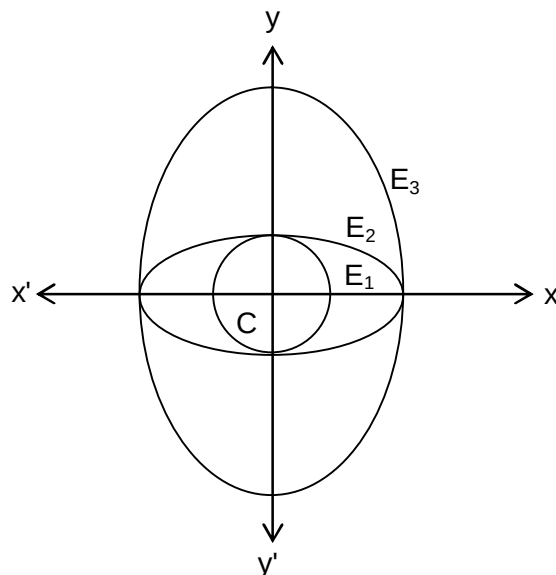
$$\therefore b_2 = 4 = b_3 e_3$$

$$\Rightarrow b_3 = \sqrt{41}$$

$$\text{and } a_3^2 = b_3^2 (1 - e_3^2)$$

$$= 41 \left(1 - \frac{16}{41} \right) = 25$$

$$\therefore \text{Ellipse } E_3 \text{ is } \frac{x^2}{25} + \frac{y^2}{41} = 1$$



47. D

48. D

Sol. (Q.47 to 48):

Let $P(h, k)$ be a point with the given property and assume $h, k > 0$

Now, without loss of generality, we take $PC > PA$

$$\text{So, } \sqrt{2}PC = PB + PD$$

$$\Rightarrow 2h^2 + 2(k+a)^2 = (h-a)^2 + (h+a)^2 + 2k^2 + 2\sqrt{(h+a)^2 + k^2} \times \sqrt{(h+a)^2 + k^2}$$

$$\Rightarrow \text{locus comes out to be } x^2 + y^2 = a^2$$

So, circumcircle of square ABCD will be locus of point P.

Section – B

49. 3

Sol. Given, $z + \omega = i$ (1)

and $z^2 + \omega^2 = 1$ (2)

$\therefore$ From (1), on squaring we get

$$z^2 + \omega^2 + 2z\omega = -1 \Rightarrow 1 + 2z\omega = -1 \quad [\text{Using (2)}]$$

$$\Rightarrow z\omega = -1 \quad \dots\dots\dots(3)$$

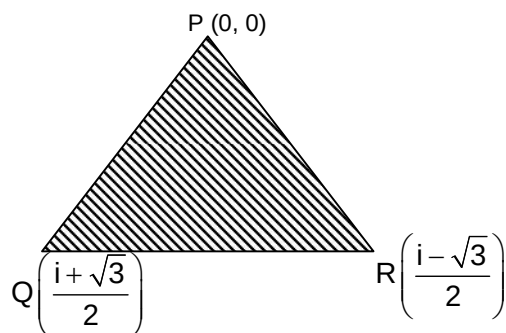
Now, let us consider a quadratic equation in x whose roots are z and ω .

$$\Rightarrow x^2 - ix - 1 = 0$$

$$\therefore x = \frac{i \pm \sqrt{i^2 + 4}}{2} = \frac{i \pm \sqrt{3}}{2}$$

Let

$$\omega = \frac{i + \sqrt{3}}{2} \quad z = \frac{i - \sqrt{3}}{2}$$



$$\text{So, ar.}(\Delta PQR) = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 1 \\ \frac{-\sqrt{3}}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2} \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \right) = \frac{\sqrt{3}}{4}$$

50. 3

Sol. $\beta = \int_0^1 \frac{2x + 3x^2 + 4x^3}{2\sqrt{x^4 + x^3 + x^2}} dx$

$$\text{Put } (x^2 + x^3 + x^4) = t \Rightarrow (2x + 3x^2 + 4x^3) dx = dt$$

$$\beta = \int_0^3 \frac{dt}{2\sqrt{t}}$$

$$\beta = \left[\sqrt{t} \right]_0^3 = \sqrt{3} \Rightarrow \beta^2 = 3$$

51. 6

Sol. $\frac{dy}{dt} = -k\sqrt{y}$

$$\frac{dy}{\sqrt{y}} = -k dt$$

$$2\sqrt{y} = -kt + c$$

$$t = 0; y = 4$$

$$4 = c$$

$$2\sqrt{y} = -kt + 4$$

$$y = 0$$

$$kt = 4$$

$$t = \frac{4}{\frac{1}{15}} = 60 \text{ minutes}$$

Section – C

52. 1.00

Sol. We have $f(x) = \begin{cases} 3x, & x \geq 0 \\ x, & x < 0 \end{cases}$ and $g(x) = \begin{cases} \frac{x}{3}, & x \geq 0 \\ x, & x < 0 \end{cases}$

Clearly f and g are inverse of each other

$$\text{Now, } h(x) = f(g(x)) = \begin{cases} 3\left(\frac{x}{3}\right) = x, & x \geq 0 \\ x, & x < 0 \end{cases}$$

$$\text{As, } h(x) = x \forall x \in \mathbb{R}$$

$$\Rightarrow k(x) = 1 + \frac{1}{\pi} (\cos^{-1} x + \cot^{-1} x)$$

Domain of $k(x) = [-1, 1]$ and $k(x)$ is decreasing function on $[-1, 1]$

As $k(x)$ is continuous function on $[-1, 1]$

$$\text{Now, } k_{\min}(x=1) = 1 + \frac{1}{\pi} (\cos^{-1} 1 + \cot^{-1} 1)$$

$$= 1 + \frac{1}{\pi} \left(0 + \frac{\pi}{4}\right) = 1 + \frac{1}{4} = \frac{5}{4}$$

$$k_{\max.}(x=-1) = 1 + \frac{1}{\pi} (\cos^{-1}(-1) + \cot^{-1}(-1))$$

$$= 1 + \frac{1}{\pi} \left[\pi + \frac{3\pi}{4}\right] = 1 + \frac{7}{4} = \frac{11}{4}$$

$$\Rightarrow \text{Range of } k(x) = \left[\frac{5}{4}, \frac{11}{4}\right]$$

53. 1.00

Sol. We have $f(x) - g(x) = (2x + |x|) - \frac{1}{3}(2x - |x|)$

$$= \frac{4x}{3} + \frac{4}{3}|x| = \begin{cases} \frac{8}{3}x; & x \geq 0 \\ 0; & x < 0 \end{cases}$$

$\therefore$ For domain of function, $0 \leq \frac{8x}{3} \leq 1$

$$\Rightarrow \text{Domain of } I(x) = \left(-\infty, \frac{3}{8}\right]$$

$$(\text{Note : Range of function } I(x) = \left[0, \frac{\pi}{2}\right])$$

54. 1.20

55. 4.00

Sol. **(Q.54 – 55):**

The equations of the bisectors are given by $x - y = 0$ and $x + y + 2 = 0$.

These bisectors intersect at the point P $(-1, -1)$

Focus S, is the foot of $\perp$ from P to MN i.e. point of intersection of lines MN

$(3x - y - 2 = 0)$ and PS $(x + 3y + 4 = 0)$.

$$\therefore \text{Focus } S \equiv \left(\frac{1}{5}, -\frac{7}{5}\right)$$

Circle passing through PMN is $(x - 0)(x - 1) + (y - 1)(y + 2) = 0$

$$\text{or } x^2 + y^2 - x + y - 2 = 0$$

Hence, equation of tangent at $(-1, -1)$, which is also the equation of directrix is $3x + y + 4 = 0$.

56. 6.00

57. 1.00

Sol. **(Q.56 – 57):**

$$\begin{aligned} & \frac{1}{10!} \{ {}^{20}C_1 + 2 \cdot {}^{10}C_3 + {}^{10}C_5 \} \\ &= \frac{{}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + {}^{10}C_7 + {}^{10}C_9}{10!} \end{aligned}$$

$$= \frac{2^9}{10!} = \frac{8^3}{(2.5)!} = \frac{8^n}{(2.c)!} \Rightarrow a = 3, b = 4, c = 5$$